

Project Name: AI-Powered Smart Hydration & Wellness Wristband

Client Name: Brad

Project Start: 3rd June, 2026

Duration: 3 weeks (Excluding Delivery time)

On your question about dimensions — I want it as slim as possible, Fitbit-sized or smaller. Something someone would actually wear on a night out without thinking about it. I'll defer to your recommendation on exact dimensions based on what's achievable with the components.

Wanted to share a quick overview of what I'm building so we can hit the ground running. The product — Umbra:

A passive behavior change wristband that reminds people to drink water. The core experience is simple — band buzzes, you drink water, that's it. No logging, no tapping, no setup required.

Two modes:

Night Mode — active during alcohol consumption, buzzes every 40 minutes default

Daily Mode — passive all day hydration, buzzes every 90 minutes default

Both adjustable in app — 20, 30, 40, 60 minute options

Non-negotiable:

The band must work completely standalone without the app. Out of the box, no pairing, no setup. Just put it on and it buzzes. The app enhances the experience but is never required for the core function.

Sensors for v1:

HR — essential

HRV calculated from HR data — no new sensor needed

Full PPG waveform capture — no new sensor, smarter firmware

Respiration rate from HR data — no new sensor needed

Skin temperature

SpO2 — already in your band

Accelerometer — movement context, steps, sleep detection, logged passively

Drop GSR for v1 — too noisy, will add in v2

The app — ruthlessly simple:

Mode selector — Daily or Night

Interval adjustment — 20, 30, 40, 60 minutes

Live HR display

Session history

Single morning check-in notification at 9am — "How do you feel 1-10?"

Battery level

Steps and sleep logged passively in background — not featured prominently in v1 UI

App aesthetic — dark background, premium feel, clean and minimal. Think Oura app meets nightlife. Not medical looking.

Form factor direction:

Aesthetic similar to Whoop, Google Fitbit Air, or Bionny band — slim, screenless, premium, all black, woven textile or quality silicone strap if possible. Something someone wears on a night out without it looking like a fitness tracker. No visible screen or pod if possible.

Scale:

One working prototype first. Then 5-10 units for beta testing once validated.

MILESTONES:

1. R&D —demo video showing sensors reading HR and SpO2, haptic firing, BLE connecting to iPhone
2. PCB —PCB complete, band assembled
3. App —iOS and Watch app delivered and connected to band
4. Final delivery - prototype arrives in New York and functions correctly

I am building a smart wristband that helps people stay hydrated while drinking alcohol. The wristband monitors biometrics in real time — heart rate, blood oxygen, and sweat levels — and delivers haptic reminders to drink water at smart intervals based on what your body is actually doing.

Think of it as a Whoop for nightlife. Not preachy, not anti-drinking — just a smart nudge that helps you feel better the next morning.

I have a working proof-of-concept app that validates the core concept. I will share this as background reference only. I want you to build this product fresh — proper architecture, clean codebase, production quality. Don't copy the reference app, just use it to understand what I'm trying to achieve.

What I Need Built

1. Wristband Hardware Prototype

Microcontroller: ESP32 or equivalent (Bluetooth LE capable)

Heart rate + SpO2 sensor: MAX30102 or equivalent

GSR sensor (galvanic skin response): sweat/hydration estimation, contact points integrated into underside of band

Haptic motor: strong enough to feel clearly on wrist in a loud environment

LiPo battery: minimum 500mAh, 6+ hour battery life

USB-C charging

Form factor: as slim as possible — Fitbit-sized or smaller. Not a dev board on a strap. 3D printed enclosure is fine for v1.

2. Wristband Firmware

Continuously read HR, SpO2, and GSR sensors

Broadcast data via Bluetooth LE to companion iPhone app

Receive haptic trigger commands from app and fire motor

Onboard fallback: if BLE drops, fire reminders independently on default interval

Low power mode between readings

Battery level reporting to app

3. iOS App (SwiftUI)

Built from scratch, production quality:

Session tracking — start/end a night out

Passive reminders — band buzzes, you drink water, no logging required

HR zone display — resting / elevated / high with color coding

Biometric dashboard — live HR, SpO2, GSR reading from wristband

Session history — log of past sessions with biometric averages

Hangover rating — next morning prompt at 9am to rate how you feel 1-10, correlated with previous night's data

Insights — over time, show what patterns lead to best and worst mornings
Design: dark, premium aesthetic — think Oura app meets nightlife. Not medical, not frat party.

4. Apple Watch Companion App (WatchOS / SwiftUI)

Mirrors active session from iPhone
Delivers haptic reminders directly on wrist
HR zone indicator
Syncs session data back to iPhone app

5. BLE Integration Layer

Swift BLE manager class for iOS handling device discovery, pairing, sensor data, and haptic commands
Documented BLE service UUIDs and characteristics
Reliable reconnection handling if connection drops mid-session

What I'm NOT Asking For

Custom PCB (breadboard/module prototype is fine for v1)
Injection molded enclosure (3D printed is fine)
App Store submission
Backend/server infrastructure

Deliverables

Working physical wristband prototype shipped to New York
All firmware source code
Full iOS app source code (Xcode project)
Apple Watch companion app source code
Swift BLE manager class + documentation
Short video demo showing band connecting to app, live HR displaying, and haptic firing on command

just adding back this message- with the updates- prob dont need GSR which was in your feasibility study: The product — Umbra:

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Session history
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A few key reminders as you get started:

GSR dropped from v1 — keep it simple and reliable for the first prototype
Band must work standalone without the app — haptic fires on fixed interval out of the box with no pairing or setup required
Night Mode default 35 minute haptic interval, Daily Mode default 90 minutes — both adjustable in app to 20, 30, 45, or 60 minutes
Haptic pattern — double tap, two short firm pulses, strong enough to feel clearly in a loud bar or festival environment
Dark premium aesthetic for the app — think Oura meets nightlife, not medical dashboard
Form factor — slim, all black, premium feel similar to Whoop or Fitbit Air
Looking forward to updates and excited to get this built together.